CDFD Runtime Paper 11: Validation, Precision, and Falsifiability

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Abstract

Executive synthesis. CDFD is useful only if it can fail. This paper defines the validation standard for the runtime series: numerical sanity checks, finite-value audits, reproducible CLI commands, experiment selection, precision analysis, and domain-specific falsification. The local experiments are treated as model diagnostics unless external validation is available.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Validation Ladder

The runtime uses a staged ladder:

1. source compiles and imports;

2. unit tests pass;
3. CDFL model validates;
4. simulation outputs are finite;
5. rerun reproduces the qualitative regime;
6. parameter sensitivity is documented;
7. independent data or experiment challenges the claim.

3 CLI Verification

The CLI pass added a reproducible command surface. A minimal verification set is:

```
python cdfd.py info
python cdfd.py domains
python cdfd.py validate examples/heat_flow.cdfl
python cdfd.py run examples/heat_flow.cdfl --out outputs/heat_flow_run.json
```

The result envelope includes provenance and `finite_audit`. Non-finite values are blockers unless the run is explicitly a boundary-failure test.

4 Test Evidence

The focused runtime check recorded:

```
PYTHONPATH=. /home/bampita/Projects/CDFD/.venv/bin/python \
  -m pytest -q tests/test_cli_runtime.py tests/test_new_engine_upgrades.py
```

with seven passing tests. These tests cover CLI runner behavior, CDFL parsing and execution, paper-aligned diagnostics, and selected native engine upgrades.

5 Precision Analysis

`experiments/reports/PRECISION_ANALYSIS.md` argues that ordinary float64 hardware is not the limiting factor for these simulations. The dominant errors are time-step selection, grid resolution, model stiffness, and CODATA measurement uncertainty for $\chi^* = \alpha^{-1} = 137.035999177$. Therefore, validation focuses on numerical convergence and model controls rather than on the laptop used to run the experiment.

6 Experiment Selection

`experiments/reports/FINAL_RELEASE_EXPERIMENT_SELECTION.md` is the current release-facing filter. It selects:

- the Mujjabi Vacuum Hysteresis Test as a candidate memory protocol;
- the Mujjabi $n = 7$ Knot Stability Test as a weak self-stabilization stress test;

- the Mujjabi Life Number Boundary Test as a bridge to Part II;
- frontier and gigamarathon sweeps as discovery coverage evidence.

It also excludes overloaded adaptive-surface output as support; that run is an instability boundary.

7 Domain Falsification

Each domain needs explicit failure conditions. For example:

- OOL: if the proposed Λ boundary is indistinguishable from an energy-only model, the tri-regime extension adds no value.
- Finance-climate: if a directed edge remains after coupling is removed, the causal result is an artifact.
- Biology/pharmacology: if the Ψ_s proxy does not stratify outcomes beyond standard variables, the mapping fails.
- Vacuum hysteresis: if residual signatures vanish after thermal and QED confounders are controlled, physical vacuum-memory interpretation fails.

8 Conclusion

Validation is not a final chapter. It is the discipline that must accompany every paper, every experiment, and every CLI output. The runtime becomes more useful by becoming more falsifiable.

References

[1] Steve Bico Mujjabi, MD. *Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant*. CDFD Part I Paper I, preprint, 2026.

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[3] Steve Bico Mujjabi, MD. *Even-n Torus Knots in CDFT: The Exclusion of $T(2,2)$, the Extension to $T(2,2m)$, and the Anti-Phase Mass Scaling Law*. CDFD Part I Paper IX, preprint, 2026.

[4] Steve Bico Mujjabi, MD. *The Vacuum Equation of State: Deriving Torus Knot Self-Consistency from the Public CDFD Balance Law $\Psi = \Phi/C$*. CDFD Part I Paper X, preprint, 2026.

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[6] Steve Bico Mujjabi, MD. *CDFD Part II: Origins of Life and Tri-Regime Bioenergetics*. Public release archive, 2026, `https://doi.org/10.5281/zenodo.20250821`.